Exact Radial-Determinant Frequencies of Annular and Circular Plates: the Closed-Ring and Solid-Disk Limits of the Sector Wave-Function Method

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Abstract

The Seok–Tiersten–Scarton wave-function method for annular-sector plates has two classical limits that never form the two-edge Galerkin determinant. At sector angle $2\Theta = 2\pi$ there are no radial walls: circumferential order is an integer n, and each frequency is a zero of a 4×4 radial edge-determinant already present in the sector assembler. At inner radius $R_i = 0$ the two origin-singular branches drop out and the same operator becomes a 2×2 at the outer radius — the classical Kirchhoff disk. Because there are no θ-edges, Vogel and Skinner's nine inner/outer combinations are different row pairs of one matrix. Frequencies are found from $\det L(n, \Omega) = 0$ from scratch; published tables enter only post hoc. A frozen sign-flip ladder replaces the sector residual bar, which does not travel here. The nine Vogel tables are recovered at 704/710 MATCH + WEAK under a frozen 1%/3% bar. The solid-disk 2×2 recovers Narita Table 2(b) at 4/4 and SP-160 Table 2.5 at 6/6; a 4×4 at $R_i = 0$ has rank three for $n \geq 1$ and is not the disk. Southwell's Table 2.31 scores 13/18; independent SHELL281 finite elements at $b/a = 0.840$, $n = 0$ give $\lambda^2 = 132.549$ against the solver's 132.990 (+0.333%) and Southwell's printed 81.0 (−63.6%). Polar-orthotropic free–free rings recover Narita Table 3 at 10/10. In-plane free–free rings recover Irie, Yamada and Muramoto's Table 2 at 48/48. Default clamped-free sector assembly is unchanged.

Keywords: annular plate; circular plate; free vibration; exact solution; radial determinant; mixed boundary conditions; polar orthotropy; in-plane vibration.

1 Introduction

Annular and circular plates are the $2\Theta = 2\pi$ and $R_i = 0$ limits of the annular-sector geometry treated in the companion paper [11]. Classical Kirchhoff tables for those limits have existed for decades: Leissa's monograph [5] compiles the exact annular results of Vogel and Skinner [6], Southwell's inner-clamped annulus as compiled in SP-160 Table 2.31 [7], and the axisymmetric work of Raju [8]; Narita [9] added five-figure free–free values for both the annulus and the solid disk; Irie, Yamada and Muramoto [10] tabulated the in-plane annular spectrum. The contribution here is not another Bessel catalogue. It is the reduction of a *validated sector solver* onto those tables, in both motions, with a screen that does not use the sector residual bar, together with all nine inner/outer edge combinations that the sector corner problem blocks, and a polar-orthotropic free–free hold-out on the same operator.

Seok, Tiersten and Scarton [1, 2, 3, 4] gave exact-wave-function solutions for annular-sector and rectangular plates, out-of-plane and in-plane, at clamped-free. The first companion paper [11] extended the annular-sector flexural assembler to completely free sectors and built a residual screen against independent finite elements; the second [12] did the rectangular analogue, where the Kirchhoff corner jump is load-bearing. Neither takes the 2π or $R_i = 0$ limit as a production entry point. That limit is cheaper than the sector: circumferential order is an integer, the characteristic matrix is 4×4 (ring) or 2×2 (disk), there are no θ-walls, and there is no corner jump. It is also the elastic forward model that a later piezoelectric identification will need on the free–free and clamped-free ring.

The detector remains paper-faithful. Frequencies are zeros of $\det L(n, \Omega) = 0$ built from the same Frobenius radial content as the sector assembler. Published values never seed a search; they enter only when a converged root is scored against them. The sign-flip ladder described in §3 was frozen before the production sweeps and was not retuned to admit a published match. Independent cyclic-symmetry SHELL281 calculations [13] are a third method, not a fitted target.

What this paper is not: McGee's solid *sector* (that is the sector assembler with a coordinate singularity); elliptical plates; simply supported or guided sector edges (the open corner-jump problem); piezoelectric coupling, mentioned here only prospectively, as future work; or a splice into either companion paper.

2 Reductions

2.1 Closed ring, out-of-plane

Requiring 2π-periodicity of the transverse displacement forces the circumferential order to an integer, $n \in \mathbb{N}_0$. For $n \geq 1$ the $\cos n\theta$ and $\sin n\theta$ solutions are degenerate and share a frequency; for $n = 0$ the mode is a singleton. Rigid-body motions on a completely free ring are $n = 0$, $s = 0$ (translation) and $n = 1$, $s = 0$ (rotation). They are excluded from the elastic tables here, as in both companion papers.

The radial operator is the 4×4 characteristic matrix already assembled for the sector at a fixed separation constant. Its rows are, in order,

$$0 = M_r(\text{inner}), \quad 1 = M_r(\text{outer}), \quad 2 = V_r(\text{inner}), \quad 3 = V_r(\text{outer}), \tag{1}$$

and searching $\det L(n, \Omega) = 0$ at fixed integer n *is* the closed ring: no θ-edge condition is weakly enforced, so no residual is generated and no residual bar is meaningful. The four Frobenius branches span $\{J_n, Y_n, I_n, K_n\}$ under the radial map used throughout the series (Supplementary Material, §S.1).

Boundary conditions enter as row swaps on (1):

- **Free (F).** Leave that edge's M_r and V_r rows in place.
- **Clamped (C).** Replace *both* of that edge's rows by W and W' evaluated at the same radius.
- **Simply supported (S).** Replace only the V_r row by W, leaving M_r.

Two conventions must be kept straight. Leissa's table titles list the *outer* edge first, while the interface used here is $(\text{bc}_{\text{inner}}, \text{bc}_{\text{outer}})$; SP-160 Table 2.22 ("Clamped, Free") is therefore inner F / outer C, and Tables 2.30 and 2.31 ("Free, Clamped") are inner C / outer F. All nine Vogel combinations are production tables below. Clamped–clamped was also checked against the thin-ring strip limit $\lambda^2 \to 22.3733\,(a/w)^2$ of width w, giving ratios 0.9999, 1.0000 and 1.0000 at b/a = 0.90, 0.94 and 0.97, before the SP-160 comparison.

2.2 Solid disk, out-of-plane

Boundedness at $r = 0$ removes Y_n and K_n. Two surviving coefficients and two outer-edge conditions give a 2×2 characteristic matrix. The practical difficulty is that the four series columns are mid-radius Taylor initial conditions rather than the Bessel pair itself, and a truncated series is finite at the origin, so "smallest $|W|$ at $r = 0$" does not isolate the regular branches. Regular solutions behave as $r^{|n|}$, which supplies two homogeneous conditions — an inner jet — whose two-dimensional kernel *is* the regular pair:

$$n = 0: \; W' = W''' = 0; \qquad n = 1: \; W = W'' = 0; \qquad n \geq 2: \; W = W' = 0. \tag{2}$$

Projecting the two outer rows of the 4×4 onto that kernel gives the production 2×2 (§S.2).

A documented failure is worth naming, because it is the natural thing to try. Evaluating the 4×4 at $R_i = 0$ and looking for its zeros gives a matrix of rank 2 at $n = 0$ but rank 3 — not 2 — for $n \geq 1$ (Table 7). Frequencies extracted from that matrix are not Leissa's disk. The production path is the 2×2. A tiny-hole 4×4 at b/a $= 10^{-3}$ is a check on that path, not the path itself, and it is a check the implementation fails on two of four rows against a pre-registered 10^{-3} relative bar (§5). That bar is not loosened.

2.3 In-plane ring

The in-plane characteristic matrix is N_r and $N_{r\theta}$ at both arcs, and the same substitution applies: fix $\xi = n$ at an integer, search Ω. There is no Kirchhoff corner term to reduce. The in-plane frequency parameter of Irie, Yamada and Muramoto is defined from the extensional rigidity $D_{\text{ext}} = Eh/(1-\nu^2)$ as $\lambda^2 = \rho h a^2 \omega^2 / D_{\text{ext}}$, that is

$$\lambda = \omega a \sqrt{\rho(1 - \nu^2)/E}, \tag{3}$$

which is a different symbol from the flexural λ^2 of (4) and must not reuse the flexural conversion factor. Their "circular" column is $\beta = b/a = 0.01$, a tiny-hole control, and is not claimed here as $R_i = 0$.

2.4 Conversion

All flexural comparisons use

$$\lambda^2 = \omega a^2 \sqrt{\rho/D}, \qquad a = R_o, \qquad D = \frac{Eh^3}{12(1 - \nu^2)}, \tag{4}$$

with h the full thickness. The map from the solver's native Ω to (4) is recomputed from the geometry and material of each run and is *not* a universal constant. On the free–free steel plate used as the fidelity anchor in [11] ($R_o = 8$ m, thickness 0.08 m, b/a $= 0.5$) the factor happens to be $4\pi^2 \approx 39.478$; at b/a $= 0.3$ it is a different number, and reusing it is an error the series has made once and now regression-tests against. The unit test is the anchor's mode 7: native $\Omega = 1.369611$ converts to $\lambda^2 = 54.0755$, matching the finite-element value to 0.01%. Native Ω is never printed with a λ^2 label. The full conversion algebra, including the separate in-plane map, is Supplementary Material, §S.3.

Poisson ratios are not mixed in a single comparison. Table 2.34 states $\nu = 1/3$ in its caption and is compared only at that value. Narita and the finite-element models use the ν printed in their own captions (0.30 or 0.33); Irie's table is $\nu = 0.3$. Vogel's comprehensive tables (2.18, 2.20, 2.22, 2.24, 2.26, 2.28, 2.30, 2.33, 2.35) print no Poisson ratio, so every sweep that touches those tables was run at both $\nu = 0.30$ and $\nu = 1/3$. The paper-facing comparison is the $\nu = 0.30$ column; those tables' ν is not asserted to be the 1/3 of Table 2.34. Table 2.16 states $\nu = 1/3$ and is compared only at that value.

3 Screen

The depth-and-gap residual bar used for completely free *sectors* [11] does not travel to this geometry: applied unchanged it rejects five genuine full-ring roots. The ring and disk therefore use a different instrument, fixed before the production sweeps were launched:

1. A coarse scan of $\log_{10}|\det L|$ on a window local to each literature target, *and* a gap-free coarse scan per n. Both, not either: the free–free wells at the anchor geometry are narrower than 5×10^{-4} in Ω, and a target-local window alone once missed a whole band.
2. Golden-section polish of each candidate minimum.
3. A real-part sign-flip ladder at $\delta = 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}$ about the polished root. CLEAN means all five inner flips are present. FLOOR means the only inner miss is at $\delta = 10^{-6}$ with all five outer flips present and the residual shrinking — the arbitrary-precision working floor at 40 digits, already named in [11]. NOT means any inner miss at $\delta \geq 10^{-5}$, or a broken outer/shrinking pattern.
4. Positive controls must come back CLEAN and negative controls must come back NOT, evaluated in the same run as the production rows.

Scoring against a published number is separate from the ladder and is applied to the relative miss alone: MATCH below 1%, WEAK below 3%, MISS otherwise. FLOOR never converts a MATCH into a MISS. The ladder is never retuned to include a published value, and the 1%/3% thresholds are not adjusted table by table. This screen is neither the persistence screen of [12] nor the sector subdomain instrument of [11]; those names do not apply here.

A search window is a hypothesis. If $\log|\det L|$ falls monotonically to an endpoint of the scanned window, the minimiser can return that endpoint. The returned value is then a property of the window, not of the plate: it is n-independent and meaningless. The sign-flip ladder catches this — every such case in this work came back NOT — but the MATCH/WEAK/MISS gate above, computed from the relative miss alone, does not. A post-hoc label (EXACT if the returned Ω is a window endpoint, NEAR if within one coarse step of one, NONE otherwise) is therefore attached to every result. The label only labels: it never moves the returned Ω, so no computed frequency changes and the solver's checkpoint version key is unaffected. An audit of all 342 stored rows of this study found 12 endpoint returns and no near-misses, every one of them in the mixed-edge sweep of §6, which is the only sweep combining a steep $\mathrm{d}\lambda^2/\mathrm{d(b/a)}$ with a fixed absolute half-window. Those cells are re-searched on a wide window where a root exists, and are marked in the tables below. Endpoint values are not frequencies and are not printed as such anywhere in this paper.

4 Out-of-plane free–free annulus

4.1 SP-160 Table 2.35

Table 2.35 of [5], from Vogel and Skinner [6], gives eight (n, s) rows at five radius ratios. Every one of the 80 cells across both Poisson ratios is MATCH or WEAK. At $\nu = 0.30$ (Table 1) all 40 cells are MATCH, the largest miss being 0.454% at $(1,1)$, b/a = 0.1, against a three-figure printed value. At $\nu = 1/3$ (Supplementary Material, §S.4) 20 cells are MATCH and 20 are WEAK, the largest miss 2.220% at $(2,1)$, b/a = 0.9. The WEAK cells at $\nu = 1/3$ cluster on the $(2,0)$ and $(3,0)$ branches, and those printed values coincide with SP-160 Table 2.34, which is Raju's $\nu = 1/3$ set [8], not Vogel's Table 2.35. Since Table 2.35 prints no Poisson ratio, the paper-facing comparison for this table is the $\nu = 0.30$ column, and Table 2.35's ν is not asserted to be the 1/3 of Table 2.34.

Table 1: Free–free annulus, $\nu = 0.30$. Printed values are SP-160 Table 2.35 [5, 6]; solver values are λ^2 from $\det L(n,\Omega) = 0$ on the 4×4 of §2.1. All 40 cells MATCH (< 1%); largest miss 0.454%. Rigid-body $n = 0, 1$ at $s = 0$ are excluded.

	b/a = 0.1		b/a = 0.3		b/a = 0.5		b/a = 0.7		b/a = 0.9	
(n,s)	printed	solver	printed	solver	printed	solver	printed	solver	printed	solver
$(2,0)$	5.30	5.3039	4.91	4.9060	4.28	4.2711	3.57	3.5725	2.94	2.9351
$(3,0)$	12.4	12.437	12.26	12.266	11.4	11.425	9.86	9.8589	8.14	8.1449
$(0,1)$	8.77	8.7747	8.36	8.3534	9.32	9.3135	13.2	13.163	34.9	34.834
$(1,1)$	20.5	20.407	18.3	18.292	17.2	17.198	22.0	21.914	55.7	55.719
$(2,1)$	34.9	34.930	33.0	32.973	31.1	31.115	37.8	37.841	93.8	93.794
$(3,1)$	53.0	52.965	51.0	51.005	47.4	47.462	55.7	55.701	135.0	135.41
$(0,2)$	38.2	38.236	50.4	50.353	92.3	92.308	251.0	250.58	2238.0	2238.9
$(1,2)$	59.0	59.072	58.8	58.784	96.3	96.266	253.0	253.13	2240.0	2240.8

4.2 Narita's five-figure hold-out

Table 2.35 is a three-figure table, so it cannot discriminate at the level the determinant actually resolves. Narita's Table 2(a) [9] at b/a = 0.3, $\nu = 0.3$ is the hold-out: four five-figure values that were not used in any development decision. All four are recovered to between 3.7×10^{-6} and 1.6×10^{-5} relative (Table 2); the printed five figures hold in every case. The $(3,0)$ row is FLOOR rather than CLEAN — its only inner miss is at $\delta = 10^{-6}$, with all five outer flips present and shrinking — which is the precision floor of the ladder, not a missing frequency.

Table 2: Narita Table 2(a) [9] five-figure hold-out, free–free annulus, $b/a = 0.3$, $\nu = 0.3$.

(n,s)	printed λ^2	solver λ^2	rel. miss	ladder
$(2,0)$	4.9060	4.9060	6.4e-06	CLEAN
$(0,1)$	8.3535	8.3535	3.7e-06	CLEAN
$(3,0)$	12.266	12.266	1.6e-05	FLOOR
$(1,1)$	18.292	18.292	1.1e-05	CLEAN

4.3 Polar-orthotropic free–free hold-out

The same 4×4 accepts a polar-orthotropic material with no new row map. Narita's three parameters $[D_\theta/D_r,\ H/D_r,\ \nu_\theta]$ map onto the radial/hoop moduli, the torsional combination T, and the reciprocal Poisson pair already used on the sector; the isotropic reduction ($E_r = E_\theta$) reproduces the isotropic $\log|\det L|$ to machine precision at $n = 0$ and $n = 2$. Table 3 of [9] is then a data-entry check. The high-modulus graphite-epoxy plate ([0.04, 0.055, 0.012]) at b/a = 0.5 and b/a = 0.3 is recovered at 10/10 MATCH (Table 3), the largest miss 4.1×10^{-4}. The one NOT ladder, $(1,1)$ at b/a = 0.3, is frequency-true at 6.9×10^{-6} and is not retuned.

Table 3: Narita Table 3 [9], high-modulus graphite epoxy $[D_\theta/D_r,\ H/D_r,\ \nu_\theta] = [0.04, 0.055, 0.012]$, free–free annulus. Narita's $\Omega = \omega a^2\sqrt{\rho h/D_r}$ is this paper's λ^2 when $D = D_r$ and h is the full thickness. All ten cells MATCH.

b/a	(n, s)	printed Ω	solver λ^2	rel. miss	ladder
0.5	$(2, 0)$	0.935	0.9348	2.22e-04	FLOOR
0.5	$(0, 1)$	1.918	1.9174	3.03e-04	FLOOR
0.5	$(3, 0)$	2.503	2.5029	2.4e-05	CLEAN
0.5	$(4, 0)$	4.636	4.6363	5.6e-05	FLOOR
0.5	$(1, 1)$	3.961	3.9614	9.7e-05	FLOOR
0.3	$(2, 0)$	1.132	1.1315	4.06e-04	CLEAN
0.3	$(0, 1)$	1.693	1.6930	1.8e-05	CLEAN
0.3	$(3, 0)$	2.897	2.8971	4.6e-05	FLOOR
0.3	$(4, 0)$	5.180	5.1795	9.6e-05	FLOOR
0.3	$(1, 1)$	4.519	4.5190	6.9e-06	NOT

4.4 Fidelity anchors and finite elements

Five free–free roots at b/a = 0.5, $\nu = 0.30$ are carried forward from [11] as a fidelity gate on the production interface rather than as a new result: the ring path must reproduce them in native Ω before any table is believed. It does, to between 2.9×10^{-6} and 1.4×10^{-5} (Table 4).

Independent cyclic-symmetry SHELL281 models at the same geometry give the elastic listing 4.2645, 9.3114, 11.4082, 17.0973, 21.0296, agreeing with the solver to between 0.023% and 0.591%. Mesh convergence at that geometry is 4.264506, 4.264461, 4.264453 for 64, 96 and 128 circumferential divisions — a relative change of 10^{-5} between the two finest meshes. Contour plots of U_Z confirm the published (n, s) assignment on every mode in the plotted set, including the $(3, 0)$, $(1, 1)$ and $(4, 0)$ rows whose labels had previously been inferred from frequency ordering alone. At b/a = 0.9 the finite-element modes 15–16 are in-plane ($|U_Z| \sim 10^{-13}$) and must not be given a flexural n; the flexural $(0, 1)$ mode at that geometry is the first axisymmetric elastic mode above that pair.

Table 4: Free–free annulus at b/a = 0.5, $\nu = 0.30$: the five fidelity anchors of [11] in native Ω, and the same modes against independent SHELL281 [13] in λ^2. Ladder FLOOR here is the 40-digit working floor, not a missing root.

(n, s)	anchor Ω	solver Ω	rel.	ladder	FE λ^2	solver λ^2	rel. (%)
$(2, 0)$	0.1081885366	0.1081900894	1.4e-05	FLOOR	4.2645	4.2711	0.155
$(0, 1)$	0.2359132157	0.2359147685	6.6e-06	FLOOR	9.3114	9.3135	0.023
$(3, 0)$	0.2894098129	0.2894082601	5.4e-06	FLOOR	11.4082	11.4254	0.151
$(1, 1)$	0.4356362253	0.4356346725	3.6e-06	FLOOR	17.0973	17.1983	0.591
$(4, 0)$	0.5336384799	0.5336400327	2.9e-06	FLOOR	21.0296	21.067	0.178

5 Out-of-plane free disk

The solid disk uses the 2×2 of §2.2. Narita's Table 2(b) [9] at $\nu = 0.33$ is again the five-figure hold-out and is recovered 4/4 at between 1.7×10^{-6} and 9.4×10^{-6} relative (Table 5). The $(2, 0)$ hold-out is MATCH on the relative miss and NOT on the sign-flip ladder; it is trusted because the finite elements sit on the same value (5.2584 against the solver's 5.2620, 0.069%). SP-160 Table 2.5 [5], six rows at the same Poisson ratio, is recovered 6/6 (Table 6); its largest disagreement is the

$(2,0)$ row, printed 5.253 against the solver's 5.262, a 0.172% difference that sits inside the 1% bar appropriate to a four-figure table and is resolved in the solver's favour by the five-figure hold-out and by the finite elements, both of which give 5.262 and 5.2584 respectively. The Table 2.5 $(0,1)$ row is likewise MATCH on error and NOT on the ladder, and is read the same way.

Independent SHELL281 models of the free disk at $\nu = 0.33$ give 5.2584, 9.0679, 12.2284 and 20.5010 for the four Narita modes: 0.068%, 0.011%, 0.13% and 0.058% from the printed values and 0.069%, 0.011%, 0.13% and 0.059% from the 2×2. Contours confirm the published (n,s) on all six Table 2.5 rows.

Table 5: Free solid disk, $\nu = 0.33$, from the 2×2 of §2.2, against Narita Table 2(b) [9] (five-figure hold-out) and independent SHELL281 [13].

(n,s)	printed λ^2	solver λ^2	rel. miss	FE λ^2	solver vs FE (%)	ladder
$(2,0)$	5.262	5.2620	2.9e-06	5.2584	0.069	NOT
$(0,1)$	9.0689	9.0689	1.7e-06	9.0679	0.011	FLOOR
$(3,0)$	12.244	12.244	9.4e-06	12.2284	0.127	CLEAN
$(1,1)$	20.513	20.513	4.7e-06	20.5010	0.059	CLEAN

Table 6: Free solid disk, $\nu = 0.33$, against SP-160 Table 2.5 [5]. All six MATCH. The $(2,0)$ row is the largest disagreement at 0.172%; the five-figure hold-out and the finite elements both sit on the solver's value.

(n,s)	printed λ^2	solver λ^2	rel. (%)	FE λ^2	ladder
$(2,0)$	5.253	5.2620	0.172	5.2584	CLEAN
$(3,0)$	12.23	12.244	0.114	12.2284	CLEAN
$(4,0)$	21.6	21.527	0.337	21.4890	CLEAN
$(0,1)$	9.084	9.0689	0.166	9.0679	NOT
$(1,1)$	20.52	20.513	0.034	20.5010	CLEAN
$(2,1)$	35.25	35.243	0.021	35.1988	CLEAN

5.1 Why the 4×4 at $R_i = 0$ is not the disk

Substituting $R_i = 0$ into the ring 4×4 and looking for its zeros is the obvious shortcut and it is wrong. Table 7 lists the ordered $\log_{10}$ singular values of that matrix at a representative frequency: at $n = 0$ there are two small values and the rank is 2, but at $n = 1, 2, 3$ only one singular value collapses and the rank is 3. A rank-3 4×4 has a one-dimensional kernel that is not the two-dimensional regular subspace, and its determinant zeros are not the plate's frequencies. This is the reason the production disk path is the 2×2 and why that fact is enforced by a regression test rather than by convention. The full regularity argument behind this rank count is Supplementary Material, §S.2.

Table 7: Ordered $\log_{10}$ singular values of the ring 4×4 evaluated at $R_i = 0$, with the observed and expected numerical rank. Rank 3 for $n \geq 1$ is why this matrix is not the disk.

	σ_1	σ_2	σ_3	σ_4	rank	expected
$n = 0$	0.43	-0.96	-16.09	-31.88	2	2
$n = 1$	0.60	-2.41	-2.88	-15.51	3	3
$n = 2$	0.60	-1.38	-3.81	-16.94	3	3
$n = 3$	0.60	-1.89	-4.91	-16.61	3	3

5.2 A named failure: the tiny-hole control

A separate control asks whether the 4×4 at a very small hole, $\mathrm{b/a} = 10^{-3}$, reproduces the 2×2 disk to a pre-registered 10^{-3} relative tolerance. It does not (Table 8): $(2,\cdot)$ and $(0,\cdot)$ differ by 0.940% and 0.545%, and the control is reported as a FAIL. The hole roots are real and deep, and both failing rows have ladder NOT while the two passing rows are CLEAN, so this is the residual physical difference between a punctured and a solid plate at that hole size, not a missed disk frequency. The tolerance is not loosened to make the control pass.

Table 8: Tiny-hole control, $\mathrm{b/a} = 10^{-3}$, $\nu = 0.33$: the 4×4 at a small hole against the production 2×2. Pre-registered bar 10^{-3} relative; the control is a named FAIL and the bar is not loosened.

mode	disk 2×2	hole 4×4	rel. (%)	ladder	control
$(2,\cdot)$	5.2620	5.3115	0.940	NOT	**FAIL**
$(0,\cdot)$	9.0689	9.1183	0.545	NOT	**FAIL**
$(3,\cdot)$	12.244	12.245	0.007	CLEAN	PASS
$(1,\cdot)$	20.513	20.524	0.054	CLEAN	PASS

6 Mixed inner and outer edges

Mixed inner/outer conditions are the content of this paper that is not a free–free echo of the companion papers, and they are reached by the row swap of §2.1 with no new derivation. Because there is no θ-edge, all nine of Vogel and Skinner's inner/outer pairs are different row pairs of one 4×4. Clamping one arc of a ring is also the mechanical boundary condition of a stator disk; that observation is a statement about the boundary condition only, and no electromechanical claim is made here.

Table 9 is the production scoreboard. Combined across both Poisson ratios the nine tables are 704/710 MATCH + WEAK. Named source skips (Table 2.30's $n = 0$ column at $\mathrm{b/a} = 0.9$; Table 2.26's printed 933; Table 2.18's two blank cells at $\mathrm{b/a} = 0.9$) are excluded from the denominators on source-internal grounds, not because the solver disagrees. The 1%/3% bar is unchanged.

Table 9: Nine Vogel inner/outer combinations as row pairs of the 4×4. Leissa titles list the outer edge first; the local interface is (inner, outer). Scores are MATCH + WEAK over both $\nu = 0.30$ and $\nu = 1/3$. Blanks and named source skips are not in the denominator.

SP-160	inner–outer	score	blanks / skips	open misses
2.35	F–F	80/80	—	—
2.18	C–C	76/76	4 blanks ($n = 2$, $\mathrm{b/a} = 0.9$)	—
2.20	S–C	80/80	—	—
2.22	F–C	80/80	—	—
2.24	C–S	80/80	—	—
2.26	S–S	78/78	printed 933 at $(2,1)$, $\mathrm{b/a} = 0.3$	—
2.28	F–S	80/80	—	—
2.30	C–F	72/76	$n = 0$ at $\mathrm{b/a} = 0.9$	$(1,0)$ at $\mathrm{b/a} = 0.1$, 0.3
2.33	S–F	78/80	—	$(1,0)$ at $\mathrm{b/a} = 0.1$
combined		704/710		

6.1 Vogel and Skinner, Tables 2.22 and 2.30

Table 2.22 (inner F, outer C) is recovered at 80/80 across both Poisson ratios and all five radius ratios including b/a = 0.9, with 78 cells MATCH and 2 WEAK (Table 10; the $\nu = 1/3$ half is §S.4). The thin-ring column is worth naming explicitly: the 4×4 does not degrade at b/a = 0.9, where the printed values run to 6183 and the solver returns 6185.1.

Table 10: Mixed edges, inner free / outer clamped, $\nu = 0.30$. Printed values are SP-160 Table 2.22 [5, 6] (Leissa's title lists the outer edge first). 80/80 MATCH + WEAK over both Poisson ratios.

	b/a = 0.1		b/a = 0.3		b/a = 0.5		b/a = 0.7		b/a = 0.9	
(n, s)	printed	solver	printed	solver	printed	solver	printed	solver	printed	solver
$(0, 0)$	10.2	10.159	11.4	11.424	17.7	17.715	43.1	43.142	360	360.35
$(1, 0)$	21.1	21.195	19.5	19.540	22.0	22.015	45.3	45.333	362	361.50
$(2, 0)$	34.5	34.535	32.5	32.594	32.0	32.115	51.5	51.585	365	364.96
$(3, 0)$	51.0	50.989	49.1	49.069	45.8	45.812	61.3	61.289	370	370.70
$(0, 1)$	39.5	39.521	51.7	51.745	93.8	93.847	253	251.64	2219	2219.5
$(1, 1)$	60.0	60.061	59.8	59.759	97.3	97.376	254	253.72	2220	2220.9
$(2, 1)$	83.4	83.478	79.0	79.061	108.0	107.49	259	259.91	2225	2225.2
$(0, 2)$	90.4	90.445	132.0	132.41	253.0	252.19	692	692.01	6183	6185.1

Table 2.30 (inner C, outer F) is reported here as 72/76 (Table 11). Four cells are excluded as one named source defect and four are direct misses. Both need justifying.

The excluded cells are the whole $n = 0$ column at b/a = 0.9, at both Poisson ratios. At that radius ratio the thin-ring $n = 0$ and $n = 1$ branches are nearly degenerate, and Vogel's *own* Table 2.22 at the same b/a prints that degeneracy: 360 against 362 for $s = 0$, and 2219 against 2220 for $s = 1$. Table 2.30 at the same b/a instead prints 51.5 against 345 and 970 against 2189. The $n = 1$ entries agree with the 4×4 to 0.06%; the $n = 0$ entries agree with nothing, and no confident root exists anywhere in $\lambda^2 \in [15, 155]$ at the first of them. The argument for excluding that column is therefore internal to the source — Vogel's other table plus the physical degeneracy — and not "the solver disagrees". The MATCH/WEAK bar is unchanged. For the record, the production sweep scored this table 72/78, excluding only the printed 51.5; the paper's denominator differs from the sweep's by treating the column as one skip rather than one cell.

The four direct misses are the $(1, 0)$ entries at b/a = 0.1 and b/a = 0.3, at both Poisson ratios: printed 3.14 against solver 3.478 (10.8%) and printed 6.33 against solver 6.552 (3.5%). They are named, not dropped, and they remain open: both lie on the $n = 1$, $s = 0$ branch at the two smallest radius ratios, where the printed values are also the smallest and coarsest in the table. Those two tables alone are 152/156; the nine-combination total is 704/710.

6.2 The axisymmetric exact values

SP-160 Table 2.16 collects exact axisymmetric frequencies at b/a = 0.5, $\nu = 1/3$, for every inner/outer pair except free–free. All eight printed cells are MATCH (Table 12), including clamped–clamped at printed 89.30 against solver 89.251 (0.055%). These are three- and four-figure printed values and the agreement is well inside the bar.

6.3 Clamped–clamped and simply supported–simply supported

Table 2.18 (inner C, outer C) is 76/76 across both Poisson ratios (Table 13; $\nu = 1/3$ is identical in λ^2 to the precision printed, as expected of a fully clamped Kirchhoff plate). The source leaves

Table 11: Mixed edges, inner clamped / outer free, $\nu = 0.30$. Printed values are SP-160 Table 2.30 [5, 6]. † the $n = 0$ column at $b/a = 0.9$ is excluded as one named source defect, on the source-internal grounds given in the text. § the production search returned its window endpoint at this cell; an endpoint is not a frequency and is not printed (see §3). The re-searched $(1, 1)$ root at $b/a = 0.1$ is WEAK at 1.37% ($\nu = 0.30$) and 1.56% ($\nu = 1/3$) and is scored WEAK, as the stored value was. Paper score 72/76.

	b/a = 0.1		b/a = 0.3		b/a = 0.5		b/a = 0.7		b/a = 0.9	
(n, s)	printed	solver	printed	solver	printed	solver	printed	solver	printed	solver
$(1, 0)$	3.14	3.4776	6.33	6.5523	13.3	13.290	37.5	37.498	345	345.17
$(0, 0)$	4.23	4.2373	6.66	6.6604	13.0	13.024	37.0	36.953	51.5†	§
$(2, 0)$	5.62	5.6202	7.96	7.9565	14.7	14.704	39.3	39.277	347	347.47
$(3, 0)$	12.4	12.448	13.27	13.276	18.5	18.562	42.6	42.655	352	351.32
$(0, 1)$	25.3	25.262	42.6	42.614	85.1	85.033	239	239.84	970†	§
$(1, 1)$	27.3	§	44.6	44.631	86.7	86.706	241	241.28	2189	2190.1
$(2, 1)$	37.0	36.943	50.9	50.947	91.7	91.738	246	245.59	2194	2193.9
$(3, 1)$	53.2	53.207	62.1	62.053	100	100.17	253	252.73	2200	2200.3

Table 12: Exact axisymmetric values, SP-160 Table 2.16 [5], b/a = 0.5, $\nu = 1/3$, $n = 0$. All eight printed combinations MATCH. Free–free is omitted in the source.

inner–outer	printed λ^2	solver λ^2	rel. (%)	ladder
C–C	89.30	89.2507	0.055	CLEAN
S–C	64.06	63.8563	0.318	CLEAN
F–C	17.51	17.5980	0.503	CLEAN
C–S	59.91	59.8987	0.019	FLOOR
S–S	40.01	40.0111	0.003	FLOOR
F–S	5.040	5.0432	0.063	FLOOR
C–F	13.05	13.0894	0.302	CLEAN
S–F	4.060	4.0700	0.247	CLEAN

$(2, 0)$ and $(2, 1)$ at b/a = 0.9 blank; those four cells (two ν) are not invented. The thin-ring column again holds: printed 2237 against 2237.2.

Table 13: Clamped–clamped annulus, $\nu = 0.30$. Printed values are SP-160 Table 2.18 [5, 6]. 76/76 MATCH + WEAK over both Poisson ratios. An em-dash marks a blank in the source; not a miss.

	b/a = 0.1		b/a = 0.3		b/a = 0.5		b/a = 0.7		b/a = 0.9	
(n, s)	printed	solver	printed	solver	printed	solver	printed	solver	printed	solver
$(0, 0)$	27.3	27.281	45.2	45.346	89.2	89.251	248	248.40	2237	2237.2
$(1, 0)$	28.4	28.766	46.6	46.644	90.2	90.230	249	249.16	2238	2237.8
$(2, 0)$	36.7	36.617	51.0	51.139	93.3	93.321	251	251.48	—	—
$(3, 0)$	51.2	51.219	60.0	60.033	99.0	98.928	256	255.44	2243	2242.7
$(0, 1)$	75.3	75.366	125	125.36	246	246.34	686	684.99	6167	6167.1
$(1, 1)$	78.6	78.635	127	127.38	248	247.74	686	686.04	6167	6167.9
$(2, 1)$	90.5	90.448	134	133.67	253	251.97	689	689.19	—	—
$(3, 1)$	112	112.11	145	144.83	259	259.16	694	694.47	6174	6174.5

Table 2.26 (inner S, outer S) is 78/78 (Table 14). The source prints 933 at $(2, 1)$, b/a = 0.3. Neighbours on that row are 71.7 and 167; a search centred on 93.3 returns $\lambda^2 = 93.371$ (CLEAN, one interior radial node). A search centred on the printed 933 returns a sixth-radial-node root at 933.60. The printed cell is a lost decimal, excluded as one named typeset at both Poisson ratios, and the MATCH/WEAK bar is not retuned to absorb 933.

Table 14: Simply supported–simply supported annulus, $\nu = 0.30$. Printed values are SP-160 Table 2.26 [5, 6]. 78/78 MATCH + WEAK over both Poisson ratios. † printed 933 is a lost decimal; the solver value is the $s = 1$ root from a search at 93.3.

	b/a = 0.1		b/a = 0.3		b/a = 0.5		b/a = 0.7		b/a = 0.9	
(n, s)	printed	solver	printed	solver	printed	solver	printed	solver	printed	solver
$(0, 0)$	14.5	14.485	21.1	21.079	40.0	40.043	110	110.06	988	987.27
$(1, 0)$	16.7	16.776	23.3	23.317	41.8	41.797	112	111.44	988	988.38
$(2, 0)$	25.9	25.936	30.2	30.273	47.1	47.089	116	115.59	993	991.70
$(3, 0)$	40.0	39.976	42.0	41.910	56.0	55.957	122	122.49	998	997.24
$(0, 1)$	51.7	51.781	81.8	81.737	159	158.64	439	439.18	3948	3948.3
$(1, 1)$	56.5	56.507	84.6	84.635	161	160.57	441	440.59	3948	3949.4
$(2, 1)$	71.7	71.686	$933^\dagger$	93.371	167	166.35	444	444.84	3952	3952.7
$(3, 1)$	94.7	94.712	108	108.20	177	176.01	453	451.92	3958	3958.2

The four combinations that mix a simply-supported edge with free or clamped (Tables 2.20, 2.24, 2.28, 2.33) are in Supplementary Material §S.4. Three of them are 80/80. Table 2.33 (inner S, outer F) is 78/80: the two misses are one printed cell at both Poisson ratios, $(1, 0)$ at b/a $= 0.1$, printed 2.30 against solver 2.409 ($\nu = 1/3$, 4.72%, CLEAN) and 2.438 ($\nu = 0.30$, 6.02%, FLOOR). That is the real nearest root, not a window endpoint, and it is the same small-b/a, $n = 1$, $s = 0$ class as the open misses of Table 2.30. The bar is not loosened; the combination is not sent to finite elements.

6.4 Southwell's Table 2.31

Southwell's values for the free-outer, clamped-inner annulus [7], compiled as SP-160 Table 2.31 at $\nu = 0.3$, are the hardest comparison in this paper and are reported as such: 13/18 under the same 1%/3% bar as every other table (Table 15). No second, table-specific metric is substituted for that score.

The split is a sensitivity, not an accuracy. Southwell's method assumed round values of the Bessel argument and inverted for the radius ratio, so his tabulated b/a is a computed quantity printed to two or three decimals while his λ^2 is exact by construction. Holding the printed λ^2 fixed and root-finding on b/a instead recovers a CLEAN, $s = 0$ root on *all eighteen* rows. Every row then carries a radius-ratio discrepancy of order 10^{-3} — including the rows that match. For the thirteen MATCH and WEAK rows that discrepancy spans 1.6×10^{-4} to 6.9×10^{-3}, a band that overlaps the misses. What separates the two groups is not the size of the discrepancy but the local sensitivity: the predicted forward error

$$\left(\frac{\mathrm{d}\lambda^2}{\mathrm{d(b/a)}} \right) \frac{|\delta(\mathrm{b/a})|}{\lambda^2} \tag{5}$$

reproduces the observed relative miss of every row to within a factor of two, 18/18, over three decades of predicted error (Supplementary Material §S.6, Fig. S.1 of that section). The five direct misses are the rows where $\mathrm{d}\lambda^2/\mathrm{d(b/a)}$ is largest.

This is a unification, and on its own it would be only an internal one. What settles the question is an independent third method.

Finite elements settle the disputed cell. The most extreme disagreement is at inner C / outer F, b/a $= 0.840$, $n = 0$, where the table prints 81.0 and the solver returns a root near 133.

Table 15: Southwell's values [7] as compiled in SP-160 Table 2.31 [5], inner clamped / outer free, $\nu = 0.3$, lowest root per n. Score 13/18 under the same bar as every other table. ‡ the production search returned its window endpoint at this cell; the value shown is from a wide re-search and is CLEAN with $s = 0$. All three re-searched rows were, and remain, MISS, so the score is unchanged.

n	b/a	printed λ^2	solver λ^2	rel. (%)	gate
0	0.276	6.25	6.2426	0.119	MATCH
0	0.642	25.0	25.745	2.981	WEAK
0	0.84	81.0	$132.99^{\ddagger}$	64.185	MISS
1	0.06	2.82	2.8148	0.184	MATCH
1	0.397	9.00	9.0207	0.230	MATCH
1	0.603	21.2	21.258	0.275	MATCH
1	0.634	25.0	25.068	0.274	MATCH
1	0.771	64.0	64.762	1.191	WEAK
1	0.827	121.0	114.17	5.642	MISS
2	0.186	6.25	6.2947	0.715	MATCH
2	0.349	9.00	9.0294	0.327	MATCH
2	0.522	16.0	16.011	0.072	MATCH
2	0.769	64.0	65.583	2.473	WEAK
2	0.81	100	96.526	3.474	MISS
3	0.43	16.0	15.785	1.342	WEAK
3	0.59	25.0	24.964	0.146	MATCH
3	0.71	49.0	$45.346^{\ddagger}$	7.457	MISS
3	0.82	100	$111.04^{\ddagger}$	11.041	MISS

A criterion was fixed before the finite-element run: a lowest FE mode within about 2% of 133 vindicates the solver; one within about 2% of 81 means the solver misses a real mode.

At C–F, b/a = 0.840, $n = 0$ the lowest flexural FE mode is $\lambda^2 = 132.549$. The solver gives 132.990 (+0.333%). Southwell prints 81.0 (−63.6%). Nothing in the 40-mode FE spectrum lies below 132.549. A refined mesh (radial divisions 16 → 24, circumferential 256 → 384) reproduces that value as 132.548973 against 132.548990, a relative change of 1.3×10^{-7}. That row is a source error, confirmed externally, and not a solver defect, a conditioning artifact, a basis truncation, or an n-labelling mistake.

The deck's positive control is the F–C, b/a = 0.5, $n = 0$ cell of Table 2.22, where the printed value is 17.7, the solver gives 17.715 and the finite elements give 17.7072: 0.041% against the printed value and 0.044% against the solver. The clamped-edge machinery is therefore correct and the remaining decks are readable.

Eighteen mixed-edge points. Table 16 is the full solver-versus-FE comparison at three C–F geometries, $n = 0$ to 5, every point sign-ladder CLEAN with $s = 0$. All eighteen agree to better than 0.8%; the solver sits *uniformly above* the finite elements, and the gap grows monotonically with n at every geometry, from +0.040% to +0.757%. That is the expected Kirchhoff-versus-Mindlin signature — the 4×4 is thin-plate while SHELL281 carries transverse shear, which softens higher-n modes more — and it is the same one-sided offset reported in [11]. Mixed-sign scatter of this magnitude would have been the worrying result; one-signed monotone growth is not. Two of Southwell's disputed rows are covered by this set: at $n = 3$, b/a = 0.820 the printed 100 is −10.5% from the FE value while the solver is +0.451%; and the MATCH control at $n = 2$, b/a = 0.522 has the printed 16.0 within 0.081% of the FE value.

A secondary differential test was also run, comparing the residual radius-ratio discrepancies of

Table 16: Solver against independent SHELL281 [13] at three inner clamped / outer free geometries, $\nu = 0.30$. Every point is sign-ladder CLEAN with $s = 0$. The offset is one-signed and grows monotonically in n: Kirchhoff versus Mindlin.

	b/a = 0.522			b/a = 0.820			b/a = 0.840		
n	FE	solver	rel. (%)	FE	solver	rel. (%)	FE	solver	rel. (%)
0	14.2651	14.2708	+0.040	104.4191	104.6938	+0.263	132.5490	132.9899	+0.333
1	14.5602	14.5707	+0.072	105.0768	105.3775	+0.286	133.2228	133.6945	+0.354
2	15.9871	16.0115	+0.153	107.0848	107.4618	+0.352	135.2700	135.8325	+0.416
3	19.7175	19.7624	+0.228	110.5422	111.0411	+0.451	138.7636	139.4725	+0.511
4	26.5058	26.5756	+0.263	115.5950	116.2545	+0.570	143.8146	144.7185	+0.629
5	36.3693	36.4717	+0.282	122.4130	123.2632	+0.695	150.5566	151.6958	+0.757

Southwell's table with those of Vogel's on the same footing. It passed a bar fixed in advance, but its own distribution is heavy-tailed and it does not control for radius ratio, which the two tables sample differently; it is reported in §S.6 as corroboration only. The finite-element result, not that test, is what carries the conclusion of this subsection.

7 In-plane free–free ring

The in-plane reduction of §2.3 is compared with Table 2 of Irie, Yamada and Muramoto [10] at $\nu = 0.3$ and $\beta = 0.2, 0.4, 0.6, 0.8$: the first two frequencies at $n = 1, 2, 3, 4$ together with the $n = 0$ radial and $n = 0$ torsional pairs, 48 cells in all. **All** 48 **are MATCH**, the largest relative miss being 0.222% at $\beta = 0.8$, $n = 2$, first mode (Table 17; the full 48-row list is §S.4). Both $n = 0$ families are recovered, which matters because the radial and torsional branches interleave differently as β grows. A handful of those MATCH cells classify NOT on the sign-flip ladder ($\beta = 0.4$ and 0.6 in Table 17); scoring is from the relative miss alone, as frozen in §3, and no in-plane finite-element check is claimed.

Comparison here is frequency-only. No in-plane eigenvector cross-check is presented and none is promised: the same comparison was attempted for rectangular in-plane modes in [12] and was not informative, for reasons that apply unchanged to this geometry. The in-plane disk is not claimed: Irie's "circular" column is $\beta = 0.01$, a tiny-hole control, and until the in-plane 2×2 is built and agrees with it, calling that column $R_i = 0$ would be an assumption rather than a result. The out-of-plane disk of §5 is a genuine $R_i = 0$ result, so the omission is specific to the in-plane motion.

Table 17: In-plane free–free ring against Irie, Yamada and Muramoto Table 2 [10], $\nu = 0.3$, with $\lambda = \omega a\sqrt{\rho(1-\nu^2)/E}$ of (3). Twelve cells per radius ratio; all 48 MATCH. "worst" names the branch carrying the largest relative miss at that β.

β	cells	gate	worst branch	worst rel. (%)	ladder classes
0.2	12	MATCH	$n = 1$	0.050	CLEAN,FLOOR
0.4	12	MATCH	$n = 2$	0.053	CLEAN,FLOOR,NOT
0.6	12	MATCH	$n = 2$	0.031	CLEAN,FLOOR,NOT
0.8	12	MATCH	$n = 2$	0.222	CLEAN,FLOOR

8 Discussion

What travels from the two companion papers is the Frobenius radial content, the conversion hygiene of §2.4, independent finite elements as a third method rather than a fitted target, and the discipline of freezing a screen before the production sweep. What does not travel is the sector residual bar — which rejects five genuine full-ring roots — the sector subdomain instrument, the rectangular persistence screen, and the Kirchhoff corner jump, which has no θ-edge to live on.

The limits of the present tables are worth stating plainly.

- Some SP-160 tables state $\nu = 1/3$ in their captions and some state no Poisson ratio at all. Every sweep that touches an unlabelled table was run at both $\nu = 0.30$ and $\nu = 1/3$, and the two are never mixed in a single comparison.
- Table 2.35 does not print $n = 4$; the finite elements carry that branch at b/a = 0.5 as $(4, 0)$ and it appears here only through the fidelity anchors.
- The $n = 0$ column of Table 2.30 at b/a = 0.9 is excluded as a source defect, on grounds internal to Vogel's own tables. The paper score is 72/76.
- The $(1, 0)$ misses of Table 2.30 at b/a = 0.1 and 0.3 are open and named.
- Southwell's Table 2.31 is 13/18 direct. The sensitivity analysis explains the split; the finite elements, not that analysis, are what resolve the extreme cell.
- The disk tiny-hole control at 10^{-3} is a named FAIL and its bar is not loosened.
- Search-window endpoint returns are labelled and excluded, never quoted as frequencies.
- Table 2.18 leaves $(2, 0)$ and $(2, 1)$ at b/a = 0.9 blank; those cells are not invented.
- Table 2.26 prints 933 at $(2, 1)$, b/a = 0.3. That is a lost decimal; the $s = 1$ root is 93.371. The printed 933 is a named typeset skip, not a MATCH.
- Table 2.33 misses $(1, 0)$ at b/a = 0.1 at both Poisson ratios (4.72% and 6.02%). Named, bar unchanged, not sent to finite elements.

9 Conclusions

A validated annular-sector solver reduces, without a new differential equation, to the classical closed ring and solid disk. The 4×4 radial determinant recovers all nine of Vogel and Skinner's inner/outer combinations at 704/710 MATCH + WEAK, Narita's polar-orthotropic free–free Table 3 at 10/10, and Irie, Yamada and Muramoto's in-plane free–free table; the 2×2 recovers the free solid disk, and the 4×4 evaluated at $R_i = 0$ demonstrably does not. Independent finite elements confirm the flexural spine at three free–free geometries, on the disk, and at three mixed-edge geometries, with a one-sided Kirchhoff-versus-Mindlin offset below 0.8% throughout — including the cell where a century-old table prints 81 and the plate vibrates at 133. Piezoelectric identification, using the free–free and clamped-free ring as its elastic forward model, is later work.

Data and code availability

The solver is released under an MIT license as a public GitHub repository, `https://github.com/McCune1/\protect\penalty\z@plate_solver`. For this paper that tree includes the ring and disk entry point (`plate_solver/ring_disk.py`), the sign-flip ladder and window-endpoint labelling, the `TestRingDisk` regression suite that keeps the disk on the 2×2 path, and the tabulated solver-versus-literature JSON under `data/paper3/`. Cluster probe scripts and mixed-edge APDL decks remain in the working package that produced the tables; they are not part of the Paper 1 or Paper 2 tags. Tables here were captured under `SOLVER_VERSION 2026-07-10.s10`, the same solver as the two companion papers; results under a different version should be checked against the clamped-free regression gates first. Printed literature values are transcribed from the sources cited and are held in one shared module, so that every table cell in this paper is either a transcription or a computed result and none is retyped by hand. The Bessel span is Supplementary Material, §S.1; disk regularity and the rank-3 trap are §S.2; the conversion algebra is §S.3; the full match lists at the second Poisson ratio and the remaining Vogel combinations are §S.4; cluster job numbers are §S.5; the Southwell inverse radius-ratio analysis is §S.6.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRediT authorship contribution statement

Garrek McCune: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Daniel Stutts:** Conceptualization, Supervision.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work, the authors used generative AI tools to assist with manuscript and code organization, drafting, editing, and formatting. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of this publication.

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